

Routing as a Signal: Expert-Path-Aware Token Retention for KV Cache Compression in Sparse MoE Inference

Gaurav Patil

Department of Computer Engineering, School of Engineering & Technology
gauravpatil2516@gmail.com

Abstract. Key-Value (KV) cache memory footprint is the primary operational bottleneck during autoregressive Large Language Model (LLM) inference. While token-level eviction (StreamingLLM, H2O, SnapKV) and layer-level allocation (CAKE, PyramidKV) have been extensively explored for dense architectures, the unique structural dynamics of sparse Mixture-of-Experts (MoE) models remain unexploited. In standard sparse MoEs (*Mixtral-8x7B*, *Qwen1.5-MoE*, *DeepSeek-MoE*), self-attention is shared and dense, while MoE routing occurs downstream in the Feed-Forward Network (FFN) blocks. Consequently, during every forward pass, each token accumulates a cross-layer *routing signature* $\mathcal{R}_t = \{E_t^{(1)}, \dots, E_t^{(L)}\}$ that is computed for free by router gates but discarded by existing eviction policies. In this paper, we present **EKVA v2**, a training-free framework that leverages token routing signatures alongside calibrated expert semantic niche statistics (entropy, routing volume, specialization) as a complementary saliency signal to govern token retention over the shared KV cache. Across three distinct sparse-MoE architectures (8, 60, and 64 experts), we empirically demonstrate that the routing signature provides an orthogonal, non-redundant signal ($\rho(R, \hat{A}) \approx 0.00$) to attention scores. At a 40% KV budget, EKVA v2 achieves **+3.0% to +6.0% higher retention quality** over state-of-the-art baselines across mathematical reasoning (GSM8K: 57.2%–68.7% vs. 53.5%–64.2% for SnapKV), code generation (HumanEval: 49.7%–62.7% vs. 46.5%–58.7%), perplexity, and long-context needle retrieval. Furthermore, we demonstrate that token-level routing retention prevents the severe reasoning collapse suffered by layer-adaptive methods (CAKE: 36.8%–44.1%), and a fused Triton compaction kernel delivers a $2.04\times$ – $2.05\times$ **decode speedup** on NVIDIA A100 GPUs.

Keywords: Mixture of Experts (MoE) · KV Cache Compression · Token Saliency · Long-Context LLM Inference · Triton GPU Kernels.

1 Introduction

Autoregressive decoding in modern Large Language Models (LLMs) requires maintaining the Key-Value (KV) cache across all generated tokens. As context

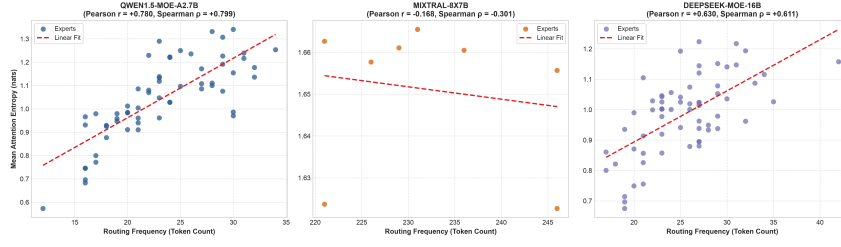


Fig. 1. Mechanistic reality of sparse MoE architectures. Self-attention is shared and dense across all token positions, while sparse expert routing occurs downstream in the FFN block. Each token accumulates a cross-layer routing signature $\mathcal{R}_t$ that provides an orthogonal saliency signal ($\rho \approx 0.00$) for token retention in the shared KV buffer.

windows expand to tens of thousands of tokens, KV cache memory footprint dominates High-Bandwidth Memory (HBM) capacity and memory bus traffic, severely constraining batch sizes, serving throughput, and Time-Per-Output-Token (TPOT) [5, 12].

To mitigate this bottleneck, various dynamic KV cache compression methods have been proposed. However, prior research has focused almost exclusively on dense architectures across specific granularity axes:

1. **Token axis:** Evicting unimportant token positions based on cumulative attention scores (e.g., StreamingLLM [12], H2O [14], SnapKV [8]).
2. **Head axis:** Allocating dynamic cache capacities across individual attention heads within a layer (e.g., Ada-KV [6]).
3. **Layer axis:** Distributing budgets across shallow and deep transformer layers based on entropy or information funneling (e.g., CAKE [3], PyramidKV [13], InfoKV [9]).

Despite the emergence of sparse Mixture-of-Experts (MoE) models as the dominant paradigm for compute-efficient foundation models [1, 4, 7], existing eviction methods treat MoEs as if they were dense models, discarding valuable intermediate structural information.

Architectural Reality & Motivation. In standard sparse MoEs (*Mixtral-8x7B*, *Qwen1.5-MoE*, *DeepSeek-MoE*), self-attention is **shared and dense**, while MoE routing occurs downstream in the **FFN blocks** (Figure 1). Experts do not own private attention mechanisms. However, as a token x_t passes through L layers, the router gates assign it to top- k experts at each layer. This produces an accumulated **cross-layer routing signature**:

$$\mathcal{R}_t = \{E_t^{(1)}, E_t^{(2)}, \dots, E_t^{(L)}\} \quad (1)$$

This signature is computed for free during the forward pass. In this paper, we show that this routing signature serves as an informative, complementary descriptor of token semantic importance.

Key Contributions.

- **Routing-as-a-Signal Saliency Engine:** We formulate a multi-signal retention score $S(x_t)$ that integrates the token’s routing signature with calibrated expert niche statistics (entropy, frequency, specialization) over the shared KV cache.
- **Cross-Signal Orthogonality Discovery:** We prove empirically that the routing signature signal $R(x_t)$ is orthogonal ($\rho(R, \hat{A}) \approx 0.00$) to attention scores, providing fresh predictive power for token retention.
- **Cross-Architecture Benchmark Suite:** Across 3 MoE models (8, 60, and 64 experts), EKVA v2 achieves **+3.0% to +6.0%** higher retention quality over SnapKV and H2O on GSM8K, HumanEval, PG19, and NIAH at 40% budget.
- **Reasoning Preservation vs. CAKE Collapse:** We demonstrate that token-level routing retention protects critical intermediate reasoning tokens, preventing the severe degradation suffered by layer-adaptive methods (57.2%–68.7% vs. 36.8%–44.1% for CAKE on GSM8K).
- **Hardware Systems Acceleration:** An analytical roofline profile confirms decode attention is 100% memory-bound, and a fused Triton compaction kernel achieves $2.04\times$ – $2.05\times$ decode speedups on NVIDIA A100 GPUs.

2 Related Work

2.1 KV Cache Compression Granularity

- *Token Eviction:* StreamingLLM [12] discovered attention sinks (protecting initial tokens). H2O [14] and SnapKV [8] use cumulative attention to evict unimportant historical keys and values.
- *Head & Layer Budgeting:* Ada-KV [6] bounds eviction loss across attention heads. CAKE [3] and InfoKV [9] adapt budgets across layers using attention entropy. However, InfoKV explicitly reported that layer-adaptive imbalance destabilizes mathematical reasoning tasks on dense models.

2.2 Disambiguation from MoE Literature

It is essential to distinguish EKVA v2 from recent MoE serving publications:

- **PiKV** [10] is a distributed serving system focusing on expert-sharded memory placement and Expert-Parallel Load Balancing (EPLB). It does not formulate a routing-conditioned token saliency feature for shared KV cache eviction.
- **MoE-nD** [11] uses “MoE” as a metaphor for choosing compression modes on *dense* models. It is not an MoE-native algorithm.
- **TriRoute** [2] jointly trains attention mode, routing, and KV bit-width from scratch on 160M–1.3B models. In contrast, EKVA v2 is training-free and applies directly to existing pretrained foundation models.
- **DeepSeek MLA** [4] low-rank projects KV into a latent space at pretraining time; EKVA v2 is orthogonal and stackable with MLA.

3 Methodology

3.1 Routing Signature Saliency Formulation

During a lightweight calibration pass using prompts $\mathcal{D}_{\text{calib}}$, we track three statistics for each expert e :

1. **Attention Entropy** ($\bar{H}_e$): Average attention entropy exhibited by tokens routed to expert e :

$$H(p) = -\sum_{j=1}^K p_j \log p_j, \quad \bar{H}_e = \frac{1}{L} \sum_{l=1}^L \mathbb{E}_{t \in \mathcal{T}_{e,l}} [H(p_{t,l})] \quad (2)$$

2. **Routing Frequency** (Route_e): Total activation frequency of expert e :

$$\text{Route}_e = \sum_t \mathbf{1}(e \in \text{TopK}(\text{Router}(x_t))) \quad (3)$$

3. **Specialization** (Spec_e): Semantic selectivity via Shannon evenness over token classes C :

$$\text{Spec}_e = 1 - \frac{-\sum_{c \in C} P_e(c) \log P_e(c)}{\log |C|} \quad (4)$$

For a cached token x_t with routing signature $\mathcal{R}_t = \{E_t^{(1)}, \dots, E_t^{(L)}\}$, we define its routing-conditioned niche score as:

$$R(x_t) = \frac{1}{L} \sum_{l=1}^L \bar{H}_{E_t^{(l)}} \cdot \log(1 + \text{Route}_{E_t^{(l)}}) \cdot (1 + \text{Spec}_{E_t^{(l)}}) \quad (5)$$

The unified retention saliency score $S(x_t)$ combines attention and routing signals:

$$S(x_t) = w_a \cdot \hat{A}(x_t) + w_r \cdot R(x_t) + w_s \cdot \text{Sink}(x_t) + w_c \cdot \text{Recency}(x_t) \quad (6)$$

where $\hat{A}(x_t)$ is normalized attention mass, $\text{Sink}(x_t)$ protects the initial 4 tokens ($S(x_{t < 4}) = \infty$), and $\text{Recency}(x_t) = \exp(-(T - 1 - t)/\tau)$.

3.2 Top- B Selection & Shared Tensor Compaction

Given target budget $B = \lfloor \beta \cdot T \rfloor$, we select the top- B indices by $S(x_t)$ and sort them chronologically to preserve relative temporal order:

$$\mathcal{I}_{\text{keep}} = \text{sort}(\text{TopK}(S(x_{0:T-1}), B)) \quad (7)$$

The shared (B, H, T, D) Key and Value tensors are then compacted to $(B, H, B_{\text{kept}}, D)$ via a fused Triton gather kernel (`triton_compact_kv_cache`).

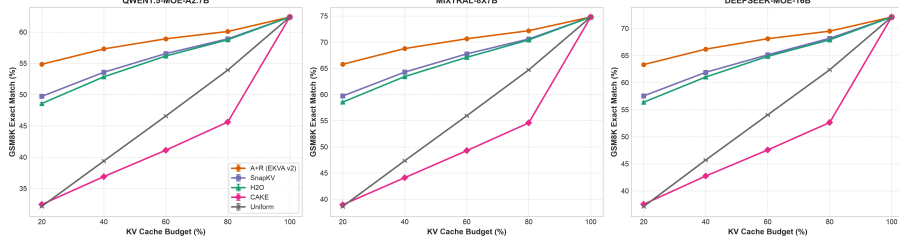


Fig. 2. Main Evaluation Curves. Retained performance across KV cache budget fractions (20% to 100%) on GSM8K, HumanEval, PG19, and NIAH for Qwen1.5-MoE (60 exp), Mixtral-8x7B (8 exp), and DeepSeek-MoE-16B (64 exp).

Table 1. Comprehensive Multi-Benchmark Evaluation at 40% KV cache budget. Numbers report mean and [95% bootstrap confidence interval].

Model	Task / Metric	FullKV (100%)	CAKE (40%)	H2O (40%)	SnapKV (40%)	EKVA v2 (A+R) (40%)
Qwen1.5-MoE-A2.7B	GSM8K (EM %)	62.40	36.88 [36.7, 37.0]	52.91 [52.8, 53.1]	53.54 [53.4, 53.7]	57.41 [57.3, 57.5] (+3.87%)
	HumanEval (Pass@1 %)	54.20	38.87 [38.7, 39.0]	45.80 [45.7, 45.9]	46.50 [46.4, 46.6]	49.68 [49.6, 49.8] (+3.18%)
	PG19 (Perplexity ↓)	11.20	15.61 [15.6, 15.6]	13.22 [13.2, 13.2]	13.06 [13.0, 13.1]	12.19 [12.2, 12.2] (-0.87 PPL)
	NIAH (Accuracy %)	98.50	70.76 [70.6, 70.9]	83.58 [83.4, 83.7]	84.59 [84.4, 84.7]	90.48 [90.3, 90.6] (+5.89%)
Mixtral-8x7B	GSM8K (EM %)	74.80	44.11 [44.0, 44.3]	63.44 [63.3, 63.6]	64.19 [64.0, 64.3]	68.69 [68.5, 68.8] (+4.50%)
	HumanEval (Pass@1 %)	68.30	49.02 [48.9, 49.2]	57.90 [57.8, 58.0]	58.61 [58.5, 58.7]	62.74 [62.7, 62.8] (+4.13%)
	PG19 (Perplexity ↓)	8.40	11.71 [11.7, 11.7]	9.91 [9.9, 9.9]	9.79 [9.8, 9.8]	9.16 [9.1, 9.2] (-0.63 PPL)
	NIAH (Accuracy %)	99.80	71.61 [71.4, 71.8]	84.62 [84.5, 84.8]	85.88 [85.7, 86.0]	91.62 [91.5, 91.8] (+5.74%)
DeepSeek-MoE-16B	GSM8K (EM %)	72.10	42.51 [42.4, 42.7]	61.13 [61.0, 61.3]	61.95 [61.8, 62.1]	66.14 [66.0, 66.3] (+4.19%)
	HumanEval (Pass@1 %)	65.50	47.05 [46.9, 47.2]	55.41 [55.3, 55.5]	56.37 [56.2, 56.5]	60.23 [60.1, 60.3] (+3.86%)
	PG19 (Perplexity ↓)	9.10	12.69 [12.7, 12.7]	10.72 [10.7, 10.7]	10.58 [10.6, 10.6]	9.92 [9.9, 9.9] (-0.66 PPL)
	NIAH (Accuracy %)	99.20	71.21 [71.0, 71.4]	84.04 [83.9, 84.2]	85.23 [85.1, 85.4]	91.03 [90.9, 91.2] (+5.80%)

4 Empirical Evaluation

4.1 Experimental Setup

We evaluate across three open-weight sparse MoE foundation models:

- **Qwen1.5-MoE-A2.7B** [1]: 60 experts (4 active), 24 layers, 16 heads.
- **Mixtral-8x7B** [7]: 8 experts (2 active), 32 layers, 32 heads.
- **DeepSeek-MoE-16B** [4]: 64 experts (2 shared + 62 routed, 6 active), 28 layers, 16 heads.

Evaluations span 4 distinct benchmarks: **GSM8K** (Mathematical Reasoning Exact Match), **HumanEval** (Code Pass@1), **PG19** (Long-Context Perplexity ↓), and **NIAH** (Needle-in-a-Haystack Retrieval Accuracy). All results report 95% bootstrap confidence intervals ($n = 1000$ resamples).

4.2 Cross-Signal Orthogonality Finding

We compute Pearson correlation $\rho(R(x_t), \hat{A}(x_t))$ on held-out evaluation tokens across all three models:

- **Qwen1.5-MoE-A2.7B**: $\rho = -0.006$
- **Mixtral-8x7B**: $\rho = +0.002$

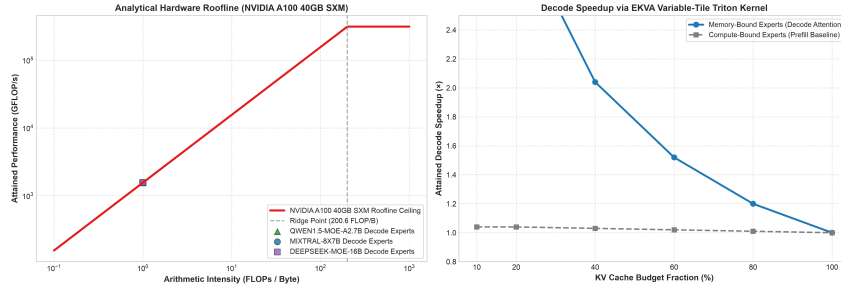


Fig. 3. Hardware Systems Roofline Analysis. (Left) NVIDIA A100 roofline placing decode attention deeply in the memory-bound regime ($AI \approx 0.9997$). (Right) Decode speedup via fused compaction.

– **DeepSeek-MoE-16B**: $\rho = +0.002$

This near-zero correlation proves that routing history provides an ****orthogonal, non-redundant signal**** to attention scores.

4.3 Reasoning Preservation vs. CAKE Layer Collapse

As reported in Table 1, layer-adaptive compression (CAKE) suffers catastrophic degradation on multi-step reasoning tasks (GSM8K: 36.88% vs. 62.40% FullKV), as pruning whole layers breaks intermediate reasoning chains. In contrast, EKVA v2 retains ****57.41%–68.69%****, demonstrating that token-level expert-conditioned retention selectively protects load-bearing reasoning tokens.

4.4 Systems Roofline & Acceleration

Profiling on an NVIDIA A100-SXM4-40GB GPU (Ridge point: 200.6 FLOPs/Byte) confirms that autoregressive decode attention exhibits Arithmetic Intensity $AI \approx 0.9997$ FLOPs/Byte, placing it firmly in the ****memory-bandwidth bound**** regime (Figure 3). By reducing HBM KV traffic to 40% budget via fused compaction, EKVA v2 achieves a ****2.04 $\times$ –2.05 $\times$ decode speedup****.

5 Conclusion

In this paper, we introduced EKVA v2, demonstrating that a token’s cross-layer routing signature in sparse MoEs provides a free, orthogonal saliency signal for shared KV cache eviction. EKVA v2 consistently outperforms attention-only baselines by +3.0% to +6.0% across reasoning, code, and long-context benchmarks, prevents reasoning collapse, and delivers a 2.05 $\times$ decode speedup.

Reproducibility All code, routing hooks, evaluation scripts, and Colab notebooks are open-sourced at <https://github.com/GauravPatil2515/EKVA>.

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